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"Be weird. Be random. Be who you are. Because you never know who would love the person you hide."

Scientific Publications

Thesis

1. Ajith Suresh. *MPCLeague: Robust MPC Platform for Privacy-Preserving Machine Learning*. PhD Thesis, 2021. Under the supervision of Prof. Arpita Patra. Indian Institute of Science (IISc), Bangalore. [PDF]
2. Ajith Suresh. *Fast Actively Secure OT Extension for Short Secrets*. Master Thesis, 2017. Under the supervision of Prof. Arpita Patra. Indian Institute of Science (IISc), Bangalore. [PDF]
3. Ajith Suresh. *Proximity-based Sentiment Analysis with Contextual Phrase Polarity*. Bachelor Thesis, 2014. College of Engineering (CET), Trivandrum.

Books ([B]) & Book Chapters ([BC])

1. [BC] Najwa Aaraj, Abdelrahaman Aly, Alvaro Garcia-Banda, Chiara Marcolla, Victor Sucasas and Ajith Suresh. *Privacy-Preserving Machine Learning for Massive IoT Deployments*. In Security and Privacy for 6G Massive IoT [PDF]

Conferences ([C]) & Journals ([J])

Publications in cryptography usually order authors alphabetically (using surnames) and conferences ([C]) are more common than journals ([J]). Workshops and affiliated events with proceedings ([W]) are marked with ‡.

1. [J] Arpita Patra, Joachim Schmidt, Thomas Schneider, Ajith Suresh and Hossein Yalame. *SynCirc: Efficient Synthesis of Depth-Optimized Circuits from High-Level Languages*. In IEEE Transactions on Computers (IEEE TC'26) (CORE rank- A*) [PDF]
2. [C] Felix Marx, Thomas Schneider, Ajith Suresh, Tobias Wehrle, Christian Weinert and Hossein Yalame. *WW-FL: Secure and Private Large-Scale Federated Learning*. In IACR Transactions on Cryptographic Hardware and Embedded Systems (CHES'26)(CORE rank- A) [PDF]
3. [C] Christopher Harth-Kitzerow, Ajith Suresh and Georg Carle. *Truncation Untangled: Scaling Fixed-Point Arithmetic for Privacy-Preserving Machine Learning to Large Models and Datasets*. In 25th Privacy Enhancing Technologies Symposium (PETS'25)(CORE rank- A) [PDF]
4. [C] Soumyadyuti Ghosh, Boyapally Harishma, Ajith Suresh, Arpita Patra, Soumyajit Dey, and Debdeep Mukhopadhyay. *Pay What You Spend! Privacy-Aware Real-Time Pricing with High Precision IEEE 754 Floating Point Division*. In 20th ACM ASIA Conference on Computer and Communications Security (ASIACCS'25) (CORE rank- A)
5. [J] Daniel Günther, Marco Holz, Benjamin Judkewitz, Helen Möllering, Benny Pinkas, Thomas Schneider and Ajith Suresh. *Privacy-Preserving Epidemiological Modeling on Mobile Graphs*. In IEEE Transactions on Information Forensics & Security (IEEE TIFS'25) (CORE rank- A) [PDF]
6. [J] Najwa Aaraj, Abdelrahaman Aly, Tim Güneysu, Chiara Marcolla, Johannes Mono, Rogerio Paludo, Iván Santos-González, Mireia Scholz, Eduardo Soria-Vazquez, Victor Sucasas and Ajith Suresh. *FANNG-MPC: Framework for Artificial Neural Networks and Generic MPC*. In IACR Transactions on Cryptographic Hardware and Embedded Systems (CHES'25)(CORE rank- A) [PDF]
7. [J] Christopher Harth-Kitzerow, Ajith Suresh, Yonqing Wang, Hossein Yalame, Georg Carle and Murali Annavaram. *High-Throughput Secure Multiparty Computation with an Honest Majority in Various Network Settings*. In 25th Privacy Enhancing Technologies Symposium (PETS'25)(CORE rank- A) [PDF] [Video]
8. [C] Yaniv Ben-Itzhak, Helen Möllering, Benny Pinkas, Thomas Schneider, Ajith Suresh, Oleksandr Tkachenko, Shay Vargaftik, Christian Weinert, Hossein Yalame and Avishay Yanai. *ScionFL: Efficient and Robust Secure Quantized Aggregation*. (Runner-Up Distinguished Paper Award) In 2nd IEEE Conference on Secure and Trustworthy Machine Learning (IEEE SaTML'24) [PDF] [Video]

9. [J] Vinod Ganapathy, Eikansh Gupta, Arpita Patra, Gokulnath Pillai and Ajith Suresh.
Privadome: Delivery Drones and Citizen Privacy.
In 24th Privacy Enhancing Technologies Symposium (PETS'24) (CORE rank- A)

10. [C] Andreas Brüggemann, Oliver Schick, Thomas Schneider, Ajith Suresh and Hossein Yalame.
Don't Eject the Impostor: Fast Three-Party Computation With a Known Cheater.
In 45th IEEE Symposium on Security and Privacy (IEEE S&P'24) (CORE rank- A*)

11. [C] Gowri R Chandran, Raine Nieminen, Thomas Schneider and Ajith Suresh.
PrivMail: A Privacy-Preserving Framework for Secure Emails.
In 28th European Symposium on Research in Computer Security (ESORICS'23) (CORE rank- A)

12. [J] Nishat Koti, Shravani Patil, Arpita Patra and Ajith Suresh.
MPClan: Protocol Suite for Privacy-Conscious Computations.
In Journal of Cryptology (JoC'23) (CORE rank- A*)

13. [C] Andreas Brüggemann, Robin Hundt, Thomas Schneider, Ajith Suresh and Hossein Yalame.
FLUTE: Fast and Secure Lookup Table Evaluations.
In 44th IEEE Symposium on Security and Privacy (IEEE S&P'23) (CORE rank- A*)

14. [W] Till Gehlhar, Felix Marx, Thomas Schneider, Ajith Suresh, Tobias Wehrle and Hossein Yalame.
SafeFL: MPC-friendly framework for Private and Robust Federated Learning[‡].
In 6th Deep Learning Security and Privacy Workshop (DLSP'23)

15. [J] Thomas Schneider, Ajith Suresh and Hossein Yalame.
Comments on "Privacy-Enhanced Federated Learning Against Poisoning Adversaries".
In IEEE Transactions on Information Forensics & Security (IEEE TIFS'23) (CORE rank- A),
In IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP'23)

Research work(s) published during PhD. I am the primary author for publications marked with [†].

16. [C] Nishat Koti, Arpita Patra, Rahul Rachuri and Ajith Suresh.
Tetrad: Actively Secure 4PC for Secure Training and Inference.[†]
In 29th Network and Distributed System Security Symposium (NDSS'22) (CORE rank- A*)

17. [C] Arpita Patra, Thomas Schneider, Ajith Suresh and Hossein Yalame.
SynCirc: Efficient Synthesis of Depth-Optimized Circuits for Secure Computation.
In IEEE International Symposium on Hardware Oriented Security and Trust (HOST'21)

18. [C] Nishat Koti, Mahak Pancholi, Arpita Patra and Ajith Suresh.
SWIFT: Super-fast and Robust Privacy-Preserving Machine Learning.[†]
In 30th USENIX Security Symposium (USENIX'21) (CORE rank- A*)

19. [C] Patra, Thomas Schneider, Ajith Suresh and Hossein Yalame.
ABY2.0: Improved Mixed-Protocol Secure Two-Party Computation.[†]
In 30th USENIX Security Symposium (USENIX'21) (CORE rank- A*)

20. [C] Arpita Patra and Ajith Suresh.
BLAZE: Blazing Fast Privacy-Preserving Machine Learning.[†]
In 27th Network and Distributed System Security Symposium (NDSS'20) (CORE rank- A*)

21. [C] Harsh Chaudhari, Rahul Rachuri and Ajith Suresh.
Trident: Efficient 4PC Framework for Privacy Preserving Machine Learning.[†]
In 27th Network and Distributed System Security Symposium (NDSS'20) (CORE rank- A*)

22. [J] Megha Byali, Harsh Chaudhari, Arpita Patra and Ajith Suresh.
FLASH: Fast and Robust Framework for Privacy-preserving Machine Learning.
In 20th Privacy Enhancing Technologies Symposium (PETS'20) (CORE rank- A)

23. [C] Harsh Chaudhari, Ashish Choudhury, Arpita Patra and Ajith Suresh.
ASTRA: High Throughput 3PC over Rings with Application to Secure Prediction.[†‡]
In ACM Conference on Cloud Computing Security Workshop (ACM CCSW'19)

Research work(s) published during M.Tech. (Research). I am the primary author for publications marked with [†].

24. [C] Arpita Patra, Pratik Sarkar and Ajith Suresh.
Fast Actively Secure OT Extension for Short Secrets.[†]
In 24th Network and Distributed System Security Symposium (NDSS'17) (CORE rank- A*)

Workshops, Symposiums & Posters

1. Martin Zbudila, Ajith Suresh, Hossein Yalame, Omid Mirzamohammadi, Aysajan Abidin, Bart Preneel.
   SoK: Connecting the Dots in Privacy-Preserving ML - Systematization of MPC Protocols and Conversions Between Secret Sharing Schemes.
   In TPMPC'26 (Contibuted Talk) [PDF]
2. Abdelrahaman Aly, Sri Harsha Gajavalli, Saurav Pawar, Eduardo Soria-Vazquez, Victor Sucasas and Ajith Suresh.
   PetalGuard: Private Federated Learning Framework for Large Language Models.
   In Flower AI Summit'25
3. Christopher Harth-Kitzerow, Ajith Suresh, Yonqing Wang, Hossein Yalame, Georg Carle and Murali Annavaram.
   High-Throughput Secure Multiparty Computation with an Honest Majority in Various Network Settings.
   In TPMPC'25 (Contibuted Talk) [PDF]
4. Soumyadyuti Ghosh, Boyapally Harishma, Ajith Suresh, Arpita Patra, Soumyajit Dey, and Debdeep Mukhopadhyay.
   Stable and Accurate Real-Time Pricing in Smart Grids.
   In TPMPC'25 (Contibuted Talk)
5. Andreas Brüggemann, Oliver Schick, Thomas Schneider, Ajith Suresh and Hossein Yalame.
   Don't Eject the Impostor - Honest-Majority MPC With Fixed Malicious Parties.
   In TPMPC'25 (Contibuted Talk) [PDF]
6. Najwa Aaraj, Abdelrahaman Aly, Tim Güneysu, Chiara Marcolla, Johannes Mono, Rogerio Paludo, Iván Santos-González, Mireia Scholz, Eduardo Soria-Vazquez, Victor Sucasas and Ajith Suresh.
   FANNG-MPC: Framework for Artificial Neural Networks and Generic MPC.
   In TPMPC'24 (Contibuted Talk) [PDF] [P] [Video]
7. Andreas Brüggemann, Thomas Schneider, Ajith Suresh and Hossein Yalame.
   Is Everyone Equally Trustworthy in Practice? (Short Talk).
   In IEEE S&P'23 (Short Talk) [Video]
8. Gowri R Chandran, Raine Nieminen, Thomas Schneider and Ajith Suresh.
   PrivMail: A Privacy-Preserving Framework for Secure Emails (Short Talk).
   In IEEE S&P'23 (Short Talk) [Video]
9. Andreas Brüggemann, Thomas Schneider, Ajith Suresh and Hossein Yalame.
   Efficient Three-Party Shuffling Using Precomputation.
   In ACM CCS'22 (Poster) [Image]
10. Daniel Günther, Marco Holz, Benjamin Judkewitz, Helen Möllering, Benny Pinkas, Thomas Schneider and Ajith Suresh.
    Privacy-Preserving Epidemiological Modeling on Mobile Graphs.
    In ACM CCS'22 (Poster)[Image] [PDF]
11. Nishat Koti, Shravani Patil, Arpita Patra and Ajith Suresh.
    MPClan: Protocol Suite for Privacy-Conscious Computations.
    In ACM CCS'22 (Poster) [Image], In NDSS'22 (Poster) [Image]
12. Ajith Suresh.
    MPCLeague: Robust MPC Platform for Privacy-Preserving Machine Learning.
    In Doctoral Symposium (AIMLSystems'22) [PDF]
13. Nishat Koti, Arpita Patra, Rahul Rachuri and Ajith Suresh.
    Tetrad: Actively Secure 4PC for Secure Training and Inference.
    In PPML'21 (ACM CCS'21) [PDF]
14. Arpita Patra, Thomas Schneider, Ajith Suresh and Hossein Yalame.
    ABY2.0: Improved Mixed-Protocol Secure Two-Party Computation.
    In PriML'21 (NeurIPS'21), In PPML'21 (ACM CCS'21), In PPML'21 (CRYPTO'21) [PDF]
15. Nishat Koti, Arpita Patra and Ajith Suresh.
    MPCLeague: Robust and Efficient Mixed-protocol Framework for 4-party Computation.
    In IEEE S&P'21 (Poster), In DPML'21 (ICLR'21) [Image] [PDF]
16. Nishat Koti, Mahak Pancholi, Arpita Patra and Ajith Suresh.
    SWIFT: Super-fast and Robust Privacy-Preserving Machine Learning.
    In ARCS'22 (Symposium), In DPML'21 (ICLR'21), In PriML/PPML'20 (NeurIPS'20) [PDF]
17. Harsh Chaudhari, Ashish Choudhury, Arpita Patra and Ajith Suresh.
    ASTRA: High Throughput 3PC over Rings with Application to Secure Prediction.
    In PPML'19 (ACM CCS'19) [PDF]

Preprints & Manuscripts

1. Martin Zbudila, Ajith Suresh, Hossein Yalame, Omid Mirzamohammadi, Aysajan Abidin, Bart Preneel.
   SoK: Connecting the Dots in Privacy-Preserving ML - Systematization of MPC Protocols and Conversions Between Secret Sharing Schemes.
   Under Submission